

Towards a Distance Vector Routing Protocol for Named Data Networking

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NDN in Mobile Adhoc Scenarios

- Current deployed NDN routing protocol (NLSR) is based on Link-state
 - Require synchronization of the link-state database across nodes
- Highly dynamic environments desire a lightweight routing protocol to propagate rapidly changing data reachability information
- Distance-vector routing protocol enables each node:
 - Selectively exchange name prefix reachability information
 - Estimate reachability to desired data in a distributed and asynchronous manner

NDN Distance Vector Routing (NDVR)

- Goal: a simplest possible way to propagate name prefix reachability information in a mobile adhoc scenario

Design:

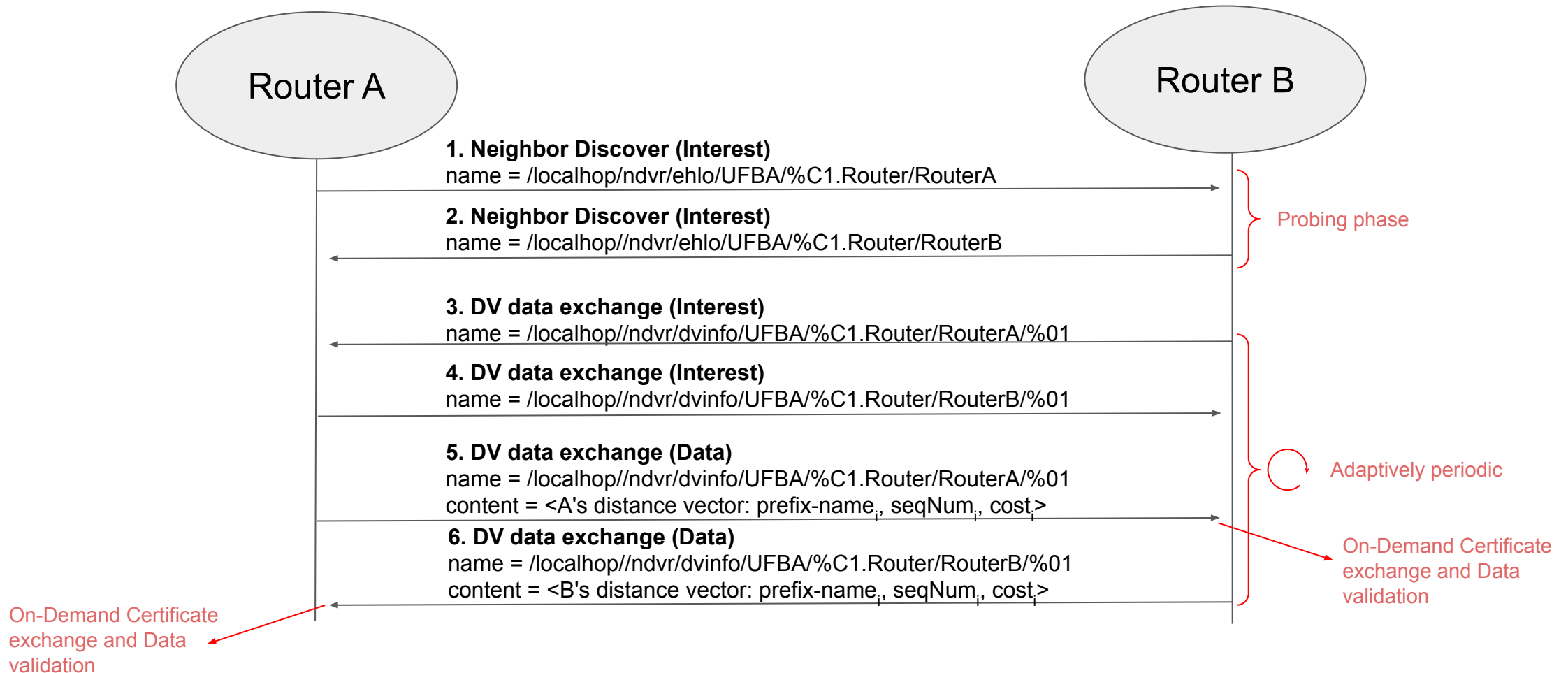
- Dynamically detect neighbors
- Upon detecting a neighbor, a node sends an Interest to retrieve the neighbor's Distance Vector Information
 - Interests are not secured
- The neighbor replies with its reachability information (Distance Vector)
 - Data packets are secure and can be validated
- Uses Distributed Bellman-Ford with Sequence Number (well-known for MANET)

NDVR namespace

- Neighbor discovery: /localhop/ndvr/ehlo/<originator-routerName>
 - This is a probing/extended-hello message (does not require Data reply)
- Distance Vector Info:
/localhop/ndvr/dvinfo/<routerName>/<version>
 - Signed by router's key
- Key exchange: <routerName>/KEY
 - <routerName> is composed by the /<networkName>/%C1.Router/<Hostname>
 - Assumption: Each node must know a trust anchor to the /<networkName>

NDVR namespace example

Note: In the example, "UFBA" is the network name



To know more, please visit us on our poster session!

Information presented on the poster session:

- Prototype developed using ndnSIM 2.8
- Preliminary evaluation in MANET scenario (data retrieval delay)
- Challenges and ongoing work..

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